

Estimating the Probability of a Conjecture to be a Theorem in PLN for Inference Control

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Artificial Intelligence and Theorem Proving 2025 (AITP-25)

Outline

- 1 Probability of Conjecture to be Theorem (in PLN)
- 2 Use such Estimates to Guide Reasoning

State of the Art (notable papers):

- *Logical Prior Probability*, Abram Demski (2016)
- *Uniform Coherence*, Scott Garrabrant et al (2016)
- *Logical Induction*, Scott Garrabrant et al (2016)

Ternary predicate relating theories, proofs and propositions

$$\Theta : \text{Theory} \times \text{Proof} \times \text{Proposition} \rightarrow \text{Bool}$$



PLN Recall

-  Non-Axiomatic Logic (NAL), *Pei Wang, 2013*
-  Probabilistic Logic Networks (PLN), *Ben Goertzel et al, 2008*
-  Subjective Logic, *Audun Jøsang, 2016*

PLN Call

Traditional Logic:

$$\Gamma \vdash T$$

$\rightarrow$

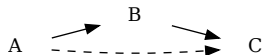
PLN:

$$\Gamma \vdash T$$

$\rightleftharpoons$

PLN Recall

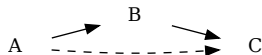
Deduction:



$$\frac{B \Rightarrow C \quad A \Rightarrow B}{A \Rightarrow C}$$

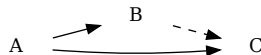
PLN Recall

Deduction:



$$\frac{B \Rightarrow C \quad A \Rightarrow B}{A \Rightarrow C}$$

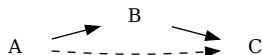
Induction:



$$\frac{A \Rightarrow C \quad A \Rightarrow B}{B \Rightarrow C}$$

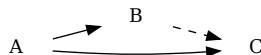
PLN Recall

Deduction:



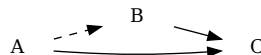
$$\frac{B \Rightarrow C \quad A \Rightarrow B}{A \Rightarrow C}$$

Induction:



$$\frac{A \Rightarrow C \quad A \Rightarrow B}{B \Rightarrow C}$$

Abduction:



$$\frac{A \Rightarrow C \quad B \Rightarrow C}{A \Rightarrow B}$$

PLN Recall

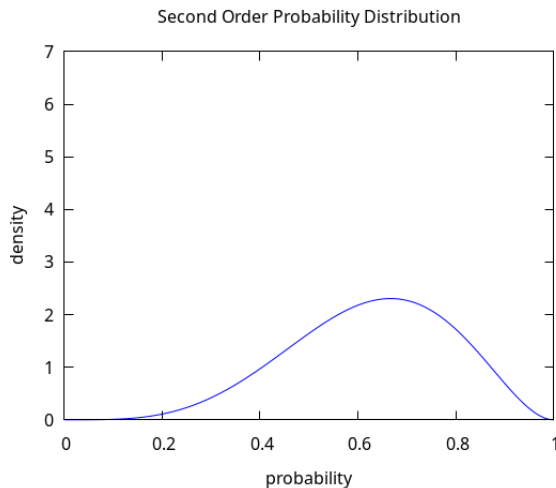
Truth Value:

$$A \Rightarrow B \stackrel{m}{=} TV$$

TV

=

*Second Order Probability
Distribution*

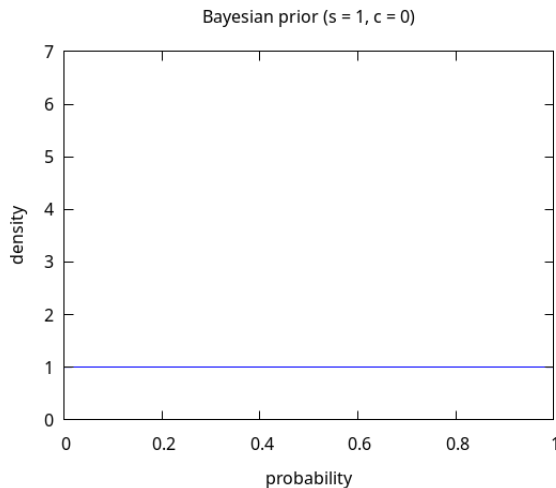


PLN Recall

Simple Truth Value:

$$A \Rightarrow B \stackrel{m}{=} \langle s, c \rangle$$

- $s = \text{strength}$
- $c = \text{confidence}$
- Beta Distribution

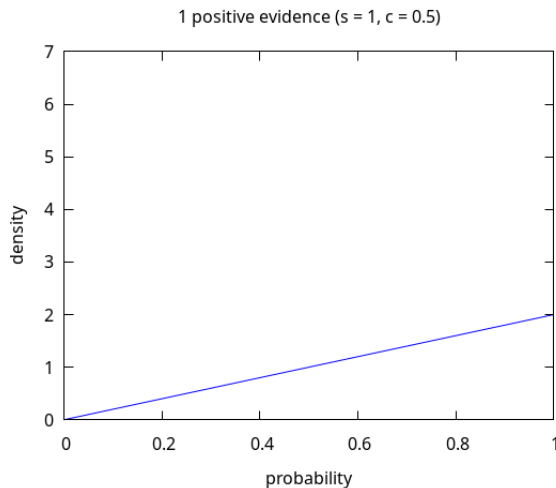


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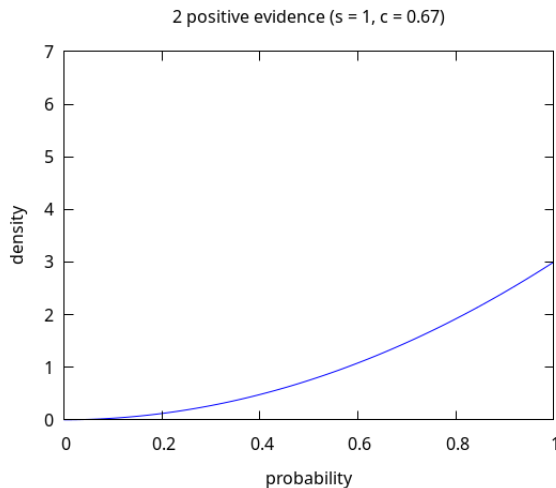


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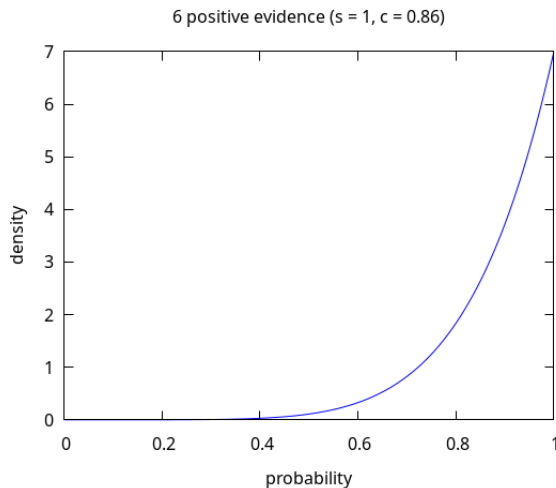


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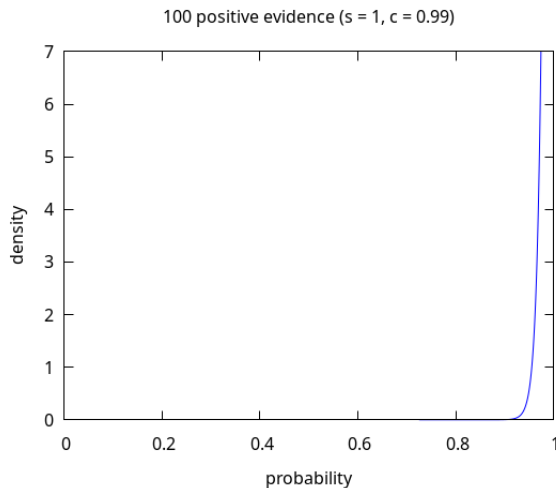


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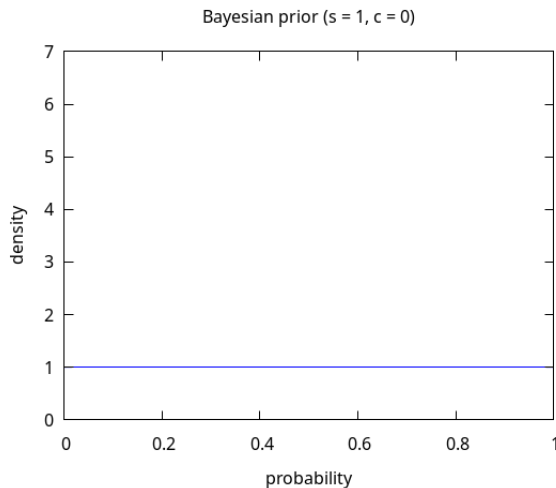


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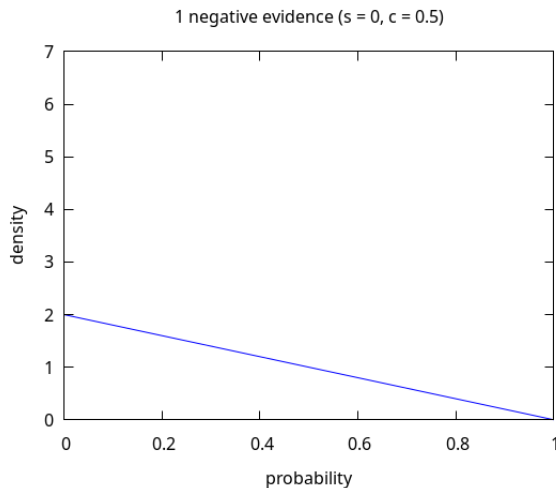


PLN Recall

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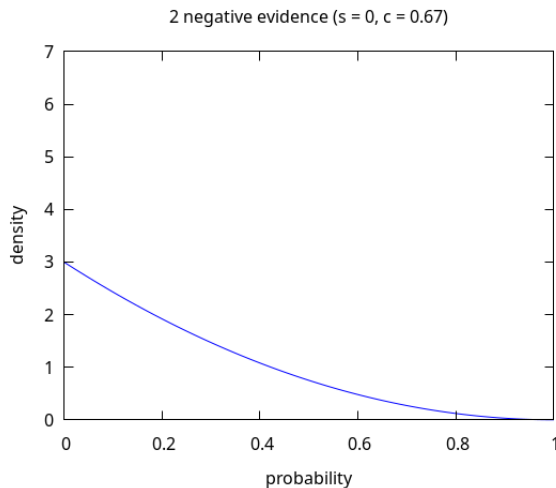


PLN Recall

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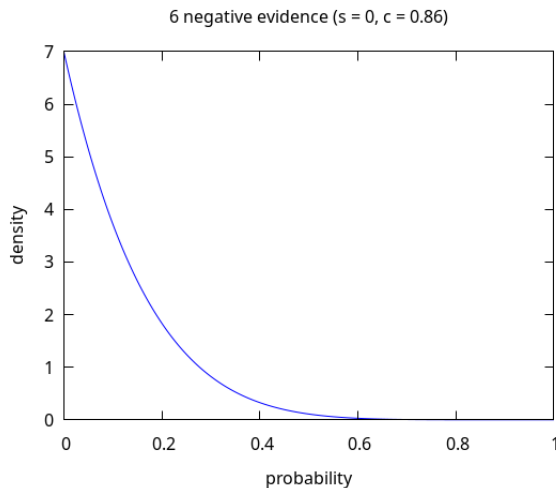


PLN Recall

Simple Truth Value:

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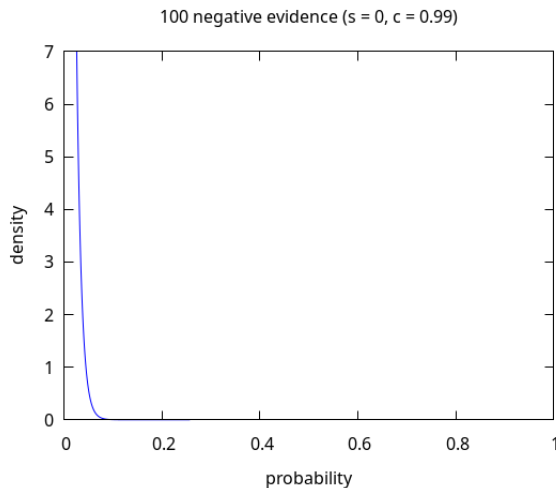


PLN Recall

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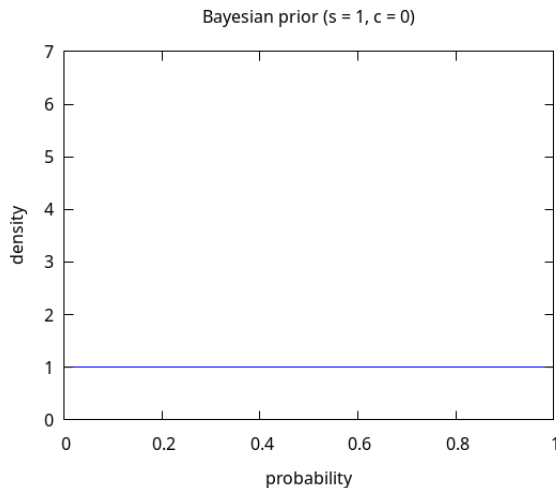


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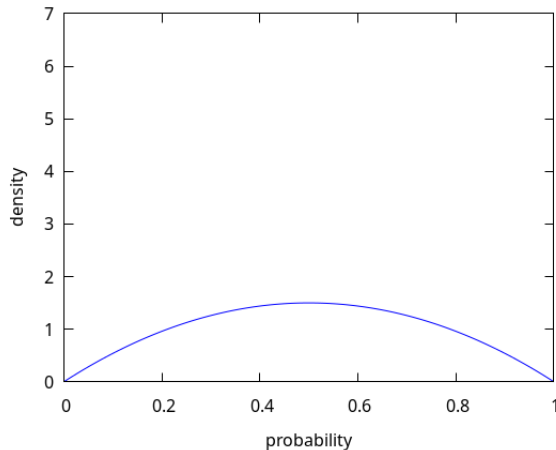
PLN Recall

Simple Truth Value:

$$A \Rightarrow B \stackrel{m}{=} \langle s, c \rangle$$

- $s = \text{strength}$
- $c = \text{confidence}$
- Beta Distribution

1 positive, 1 negative evidence ($s = 0.5, c = 0.67$)



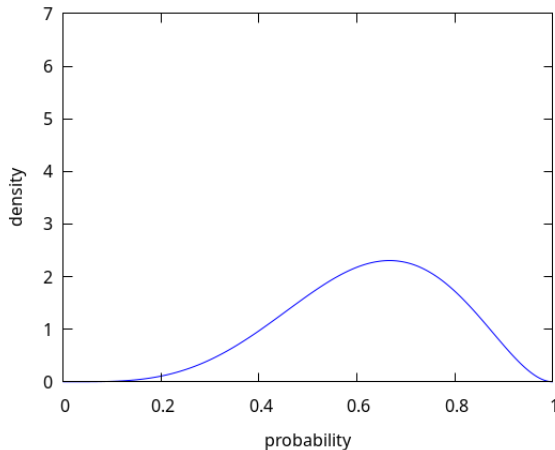
PLN Recall

Simple Truth Value:

$$A \Rightarrow B \stackrel{m}{=} \langle s, c \rangle$$

- $s = \text{strength}$
- $c = \text{confidence}$
- Beta Distribution

4 positive, 2 negative evidence ($s = 0.67, c = 0.86$)



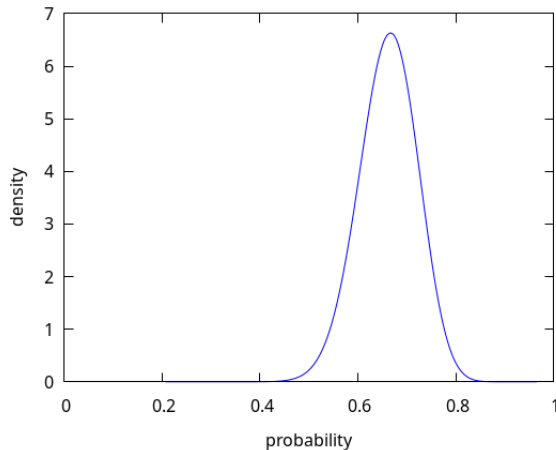
PLN Recall

Simple Truth Value:

$$A \Rightarrow B \stackrel{m}{=} \langle s, c \rangle$$

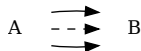
- $s = \text{strength}$
- $c = \text{confidence}$
- Beta Distribution

40 positive, 20 negative evidence ($s = 0.67$, $c = 0.98$)

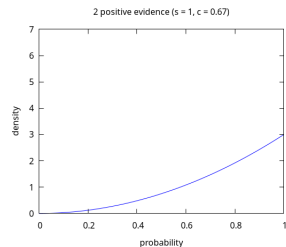
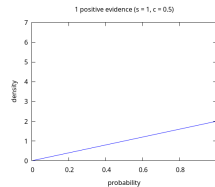
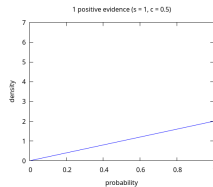


PLN Recall

Revision:

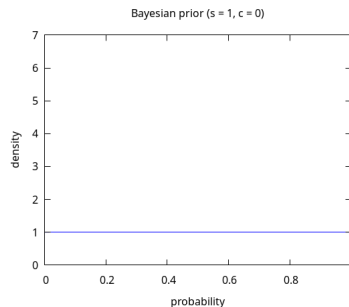
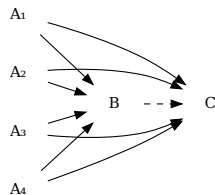


$$\frac{\overline{A \Rightarrow B} (e) \quad \overline{A \Rightarrow B} (f) \quad e \perp f}{A \Rightarrow B}$$



PLN Recall

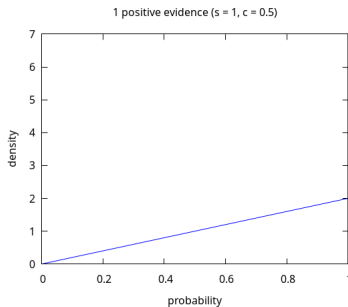
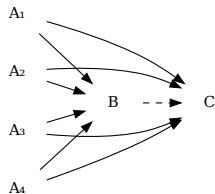
Induction + Revision:



$$\begin{array}{c}
 \frac{A_1 \Rightarrow C \quad A_1 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_2 \Rightarrow C \quad A_2 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_3 \Rightarrow C \quad A_3 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_4 \Rightarrow C \quad A_4 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \\
 \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)}
 \end{array}$$

PLN Recall

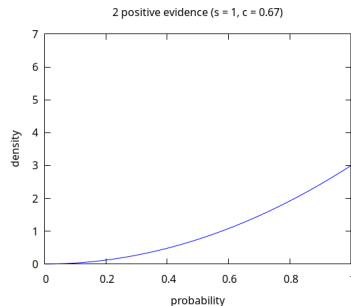
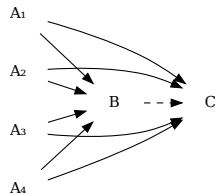
Induction + Revision:



$$\begin{array}{c}
 \frac{A_1 \Rightarrow C \quad A_1 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_2 \Rightarrow C \quad A_2 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \\
 \frac{B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{A_3 \Rightarrow C \quad A_3 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_4 \Rightarrow C \quad A_4 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \\
 \frac{B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)}
 \end{array}$$

PLN Recall

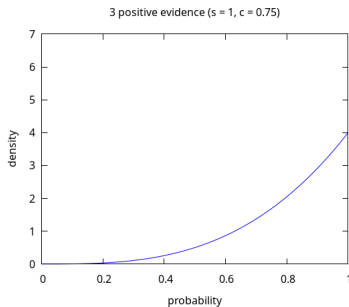
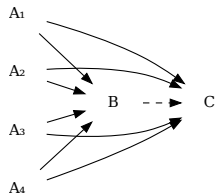
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 \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{A_3 \Rightarrow C \quad A_3 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \\
 \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{A_4 \Rightarrow C \quad A_4 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \\
 \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)}
 \end{array}$$

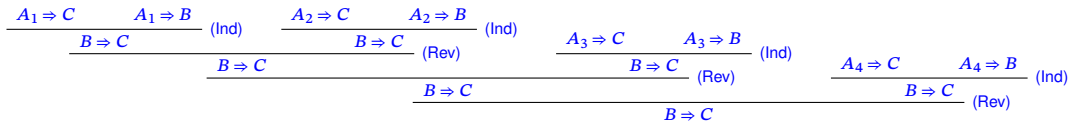
PLN Recall

Induction + Revision:



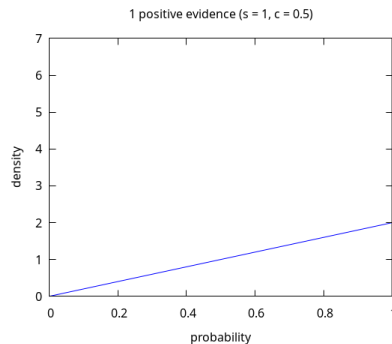
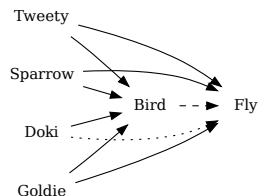
$$\begin{array}{c}
 \frac{A_1 \Rightarrow C \quad A_1 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_2 \Rightarrow C \quad A_2 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_3 \Rightarrow C \quad A_3 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \quad \frac{A_4 \Rightarrow C \quad A_4 \Rightarrow B}{B \Rightarrow C} \text{ (Ind)} \\
 \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)} \quad \frac{B \Rightarrow C \quad B \Rightarrow C}{B \Rightarrow C} \text{ (Rev)}
 \end{array}$$

The diagram shows four input variables labeled A1, A2, A3, and A4 on the left. Arrows from each Ai point towards two central nodes, B and C. A dashed arrow points from B to C, representing a causal relationship between them.



PLN Recall

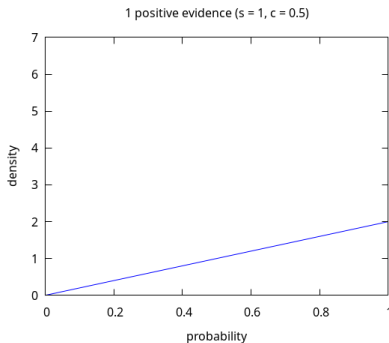
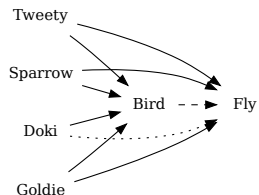
Induction + Revision example:



$$\frac{\text{Tweety} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 1 \rangle \quad \text{Tweety} \Rightarrow \text{Bird} \stackrel{m}{=} \langle 1, 1 \rangle}{\text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 0.5 \rangle} \quad (\text{Ind}, t)$$

PLN Recall

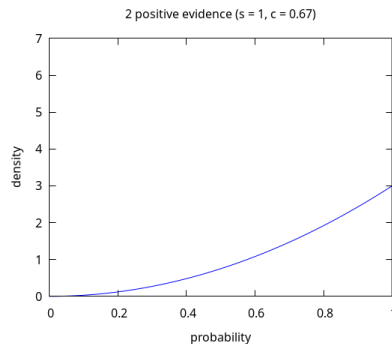
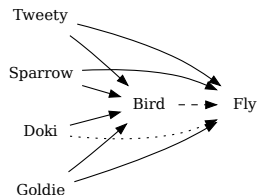
Induction + Revision example:



$$\frac{\text{Sparrow} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 1 \rangle \quad \text{Sparrow} \Rightarrow \text{Bird} \stackrel{m}{=} \langle 1, 1 \rangle}{\text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 0.5 \rangle} \quad (\text{Ind}, s)$$

PLN Recall

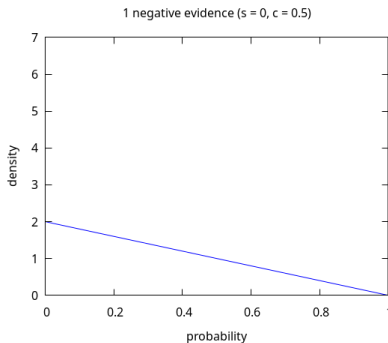
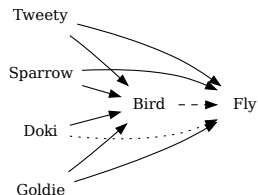
Induction + Revision example:



$$\frac{\text{Bird} \Rightarrow \text{Fly} \stackrel{\text{m}}{=} \langle 1, 0.5 \rangle \quad (t) \quad \text{Bird} \Rightarrow \text{Fly} \stackrel{\text{m}}{=} \langle 1, 0.5 \rangle \quad (s)}{\text{Bird} \Rightarrow \text{Fly} \stackrel{\text{m}}{=} \langle 1, 0.67 \rangle} \quad t \perp s \quad (\text{Rev}, t, s)$$

PLN Recall

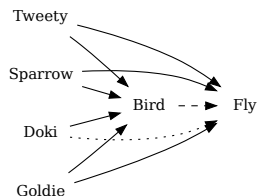
Induction + Revision example:



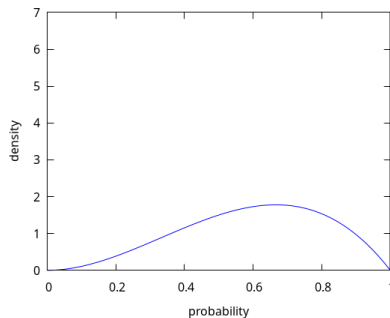
$$\frac{\text{Doki} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 0, 1 \rangle \quad \text{Doki} \Rightarrow \text{Bird} \stackrel{m}{=} \langle 1, 1 \rangle}{\text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 0, 0.5 \rangle} \quad (\text{Ind}, d)$$

PLN Recall

Induction + Revision example:



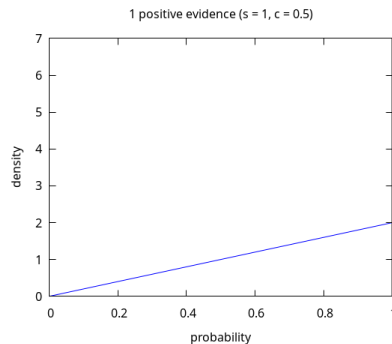
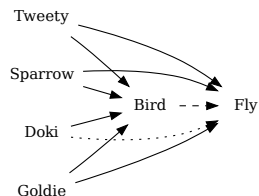
2 positive, 1 negative evidence ($s = 0.67$, $c = 0.75$)



$$\frac{\text{Bird} \Rightarrow \text{Fly} \stackrel{\text{m}}{=} \langle 1, 0.67 \rangle \quad (t,s) \quad \text{Bird} \Rightarrow \text{Fly} \stackrel{\text{m}}{=} \langle 0, 0.5 \rangle \quad (d) \quad t, s \perp d}{\text{Bird} \Rightarrow \text{Fly} \stackrel{\text{m}}{=} \langle 0.67, 0.75 \rangle} \quad (\text{Rev}, t, s, d)$$

PLN Recall

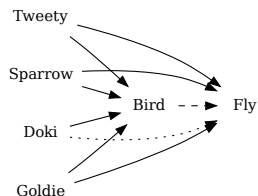
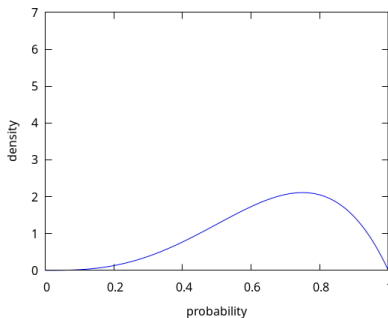
Induction + Revision example:



$$\frac{\text{Goldie} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 1 \rangle \quad \text{Goldie} \Rightarrow \text{Bird} \stackrel{m}{=} \langle 1, 1 \rangle}{\text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 0.5 \rangle} \quad (\text{Ind}, g)$$

PLN Recall

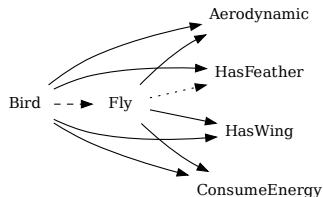
Induction + Revision example:

3 positive, 1 negative evidence ($s = 0.75$, $c = 0.8$)

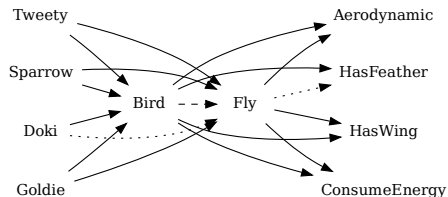
$$\frac{\text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 0.67, 0.75 \rangle \quad (t,s,d) \quad \text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 1, 0.5 \rangle \quad (g)}{\text{Bird} \Rightarrow \text{Fly} \stackrel{m}{=} \langle 0.75, 0.8 \rangle} \quad t, s, d \perp g \quad (\text{Rev}, t, s, d, g)$$

PLN Recall

Abduction + Revision:



Induction + Abduction + Revision:



PLN also has:

- Quantifiers $\exists, \forall$
- Traditional Connectors $\wedge, \vee, \neg$
- Probabilistic Computational Model

PLN and Theorem Proving

Uncertain Reasoning:

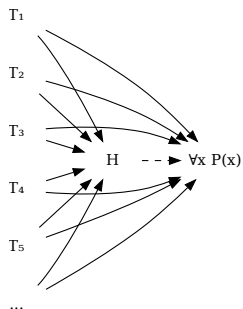
$$H \Rightarrow \forall x P(x) \text{ ?}$$

PLN and Theorem Proving

Uncertain Reasoning:

$$H \Rightarrow \forall x P(x) \text{ ?}$$

Induction + Revision:

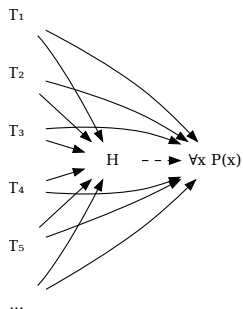


PLN and Theorem Proving

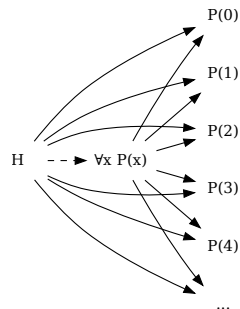
Uncertain Reasoning:

$$H \Rightarrow \forall x P(x) \text{ ?}$$

Induction + Revision:



Abduction + Revision:

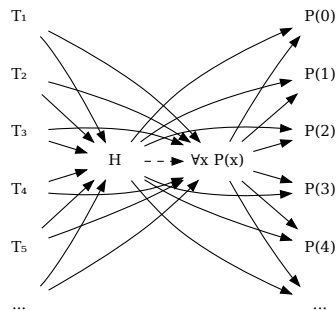


PLN and Theorem Proving

Uncertain Reasoning:

$$H \Rightarrow \forall x P(x) \text{ ?}$$

Induction + Abduction + Revision:



Ternary predicate relating theories, proofs and propositions:

$$\Theta : \text{Theory} \times \text{Proof} \times \text{Proposition} \rightarrow \text{Bool}$$

- **Theory**: *Typing relationships* encoding axioms and inference rules.

$$\{Z : \text{Nat}, S : \text{Nat} \rightarrow \text{Nat}\}$$

- **Proof**: *Inhabitant* of a type.

$$(S \ (S \ (S \ Z)))$$

- **Proposition**: *Type*.

$$\text{Nat}$$

$$\Theta(\{Z : \text{Nat}, S : \text{Nat} \rightarrow \text{Nat}\}, (S \ (S \ (S \ Z))), \text{Nat}) \stackrel{m}{=} \langle 1, 1 \rangle$$

Example (Propositional Calculus):

Instance:

$$\Theta \left(\left\{ \begin{array}{l} \text{ax-1} : (\phi \rightarrow (\psi \rightarrow \phi)), \\ \text{ax-2} : ((\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))), \\ \text{ax-3} : (((\neg \phi) \rightarrow (\neg \psi)) \rightarrow (\psi \rightarrow \phi)), \\ \text{ax-mp} : \phi \rightarrow (\phi \rightarrow \psi) \rightarrow \psi \end{array} \right\}, \right. \\ \left. (\lambda \text{ mp2.1 mp2.3 } (\text{ax-mp mp2.1 mp2.3})), \right. \\ \left. \phi \rightarrow (\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow (\psi \rightarrow \chi) \right) \stackrel{\text{m}}{=} \langle 1, 1 \rangle$$

Example (Propositional Calculus):

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$$\Theta \left(\left\{ \begin{array}{l} \text{ax-1} : (\phi \rightarrow (\psi \rightarrow \phi)), \\ \text{ax-2} : ((\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))), \\ \text{ax-3} : (((\neg \phi) \rightarrow (\neg \psi)) \rightarrow (\psi \rightarrow \phi)), \\ \text{ax-mp} : \phi \rightarrow (\phi \rightarrow \psi) \rightarrow \psi \end{array} \right\}, \right. \\ \left. (\lambda \text{ mp2.1 mp2.3 } (\text{ax-mp mp2.3 mp2.1})), \right. \\ \left. \phi \rightarrow (\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow (\psi \rightarrow \chi) \right) \stackrel{\text{m}}{=} \langle 0, 1 \rangle$$

PLN and Theorem Proving

Crisp Patterns:

- Modus ponens:

$$\Theta(\Gamma, f, a \rightarrow b) \wedge \Theta(\Gamma, x, a) \Rightarrow \Theta(\Gamma, f(x), b)$$

$$\underline{\underline{m}}$$

$$\langle 1, 1 \rangle$$

- Existential Quantification Introduction:

$$\Theta(\Gamma, \Pi, T) \wedge (\text{cl } \Gamma) \wedge (\text{cl } \Pi) \wedge (\text{cl } T) \Rightarrow \exists \pi \Theta(\Gamma, \pi, T)$$

$$\underline{\underline{m}}$$

$$\langle 1, 1 \rangle$$

PLN and Theorem Proving

Uncertain Patterns:

NEXT

Θ , Probabilistic Logic Networks (PLN)

- How likely is there a *proof* of T in Γ :

$$\exists \pi \Theta(\Gamma, \pi, T) \stackrel{m}{=} \$TV$$

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- How likely is there a *proof* of a *theorem* in a *theory* with certain *properties*:

$$\exists \gamma, \pi, \tau \Theta(\gamma, \pi, \tau) \wedge P(\gamma) \wedge Q(\pi) \wedge R(\tau) \wedge S(\gamma, \pi, \tau) \stackrel{m}{=} \$TV$$

Estimate Probability of Conjecture to be Theorem in Practice

```
(bc PARAMS THEORY (: $proof PROP))
```

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```
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```

↓ $[\$proof] = z$

```
(bc PLN_PARAMS PLN_THEORY (: $pln_proof ( $\stackrel{m}{=}$  ( $\exists$  z ( $\Theta$  [THEORY] z [PROP]))) $TV))
```

Estimate Probability of Conjecture to be Theorem in Practice

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```

↓

$\$TV = ?$

Estimate Probability of Conjecture to be Theorem in Practice

$$(\text{bc PARAMS THEORY } (: \$\text{proof PROP}))$$

$$\downarrow \quad [\$proof] = z$$

$$(\text{bc PLN_PARAMS PLN_THEORY } (: \$\text{pln_proof } (\stackrel{\text{m}}{=} (\exists z \ (\Theta \text{ [THEORY]} z \text{ [PROP]}))) \ \$TV))$$

$$\downarrow$$

$$\$TV = ?$$

- Complete ignorance: $\$TV = \langle 1, 0 \rangle$

Estimate Probability of Conjecture to be Theorem in Practice

$$(\text{bc } \text{PARAMS } \text{THEORY } (: \$\text{proof } \text{PROP}))$$

$$\downarrow \quad [\$proof] = z$$

$$(\text{bc } \text{PLN_PARAMS } \text{PLN_THEORY } (: \$\text{pln_proof } (\overset{\text{m}}{=} (\exists z \ (\Theta [\text{THEORY}] z [\text{PROP}]))) \$\text{TV}))$$

$$\downarrow$$

$$\$TV = ?$$

- Complete ignorance: $\$TV = \langle 1, 0 \rangle$
- Complete certainty: $\$TV = \langle 1, 1 \rangle$

Estimate Probability of Conjecture to be Theorem in Practice

$$(\text{bc PARAMS THEORY } (: \$\text{proof PROP}))$$

$$\downarrow \quad [\$proof] = z$$

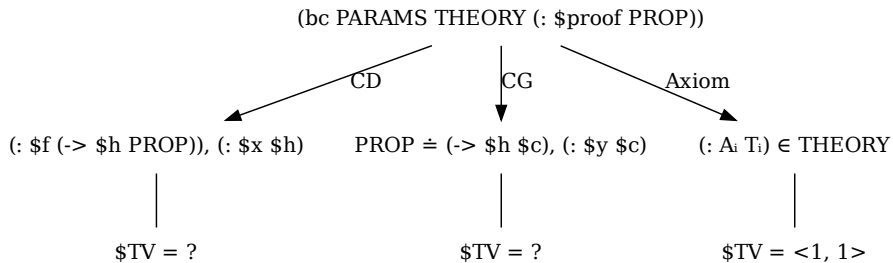
$$(\text{bc PLN_PARAMS PLN_THEORY } (: \$\text{pln_proof } (\stackrel{\text{m}}{=} (\exists z \ (\Theta \text{ [THEORY]} z \text{ [PROP]}))) \ \$TV))$$

$$\downarrow$$

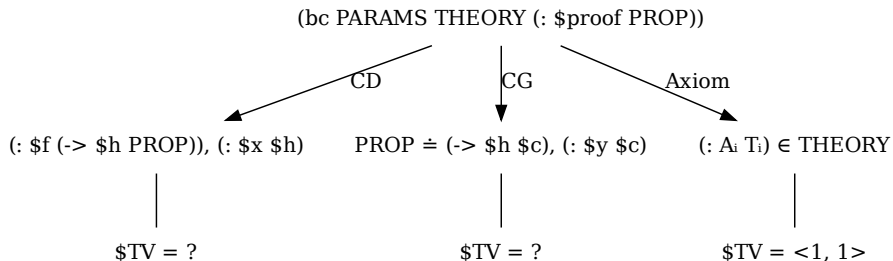
$$\$TV = ?$$

- Complete ignorance: $\$TV = \langle 1, 0 \rangle$
- Complete certainty: $\$TV = \langle 1, 1 \rangle$
- Partial certainty: $\$TV = \langle 0.7, 0.8 \rangle$

Estimate Probability of Conjecture to be Theorem as Guide



Estimate Probability of Conjecture to be Theorem as Guide



```
(bc  PLN_PARAMS PLN_THEORY
  (: $pln_proof ( $\equiv$  ( $\exists$  f h x ( $\wedge$  ( $\ominus$  [THEORY] f (-> h [PROP]))
                                     ( $\ominus$  [THEORY] x h)) $TV)))
```

Conclusion

- Early experiment (<https://github.com/trueagi-io/chaining/tree/main/experimental/pln-inf-ctl>)

- Small MetaMath corpus
- 1th run: exhaustive search, *populate* Θ

```
(≡ (Θ (Cons ([:] [a1i.1] [φ]) [PC]) [a1i.1] [φ]) (STV 1 1))
(≡ (Θ [PC] [ax-3] ([→] ([→] ([¬] [φ]) ([¬] [ψ]))) ([→] [ψ] [φ]))) (TV 1 1))
(≡ (Θ (Cons ([:] [a1i.1] [φ]) [PC]) [ax-1] ([→] [φ] ([→] [ψ] [φ]))) (STV 1 1))
(≡ (Θ (Cons ([:] [a1i.1] [φ]) [PC]) ([ax-mp] [a1i.1] [ax-1]) ([→] [ψ] [φ]))) (STV 1 1))
...
```

- 2nd run: speed-up via *PLN existential reasoning*
- Glorified Memoizer
- Future work
 - Large MetaMath corpus
 - Induction, abduction and more